

Truth:

0/0 of the True (the shortest description of the world)

Michael Grafiel S Puno

2026

The Removable Singularity Framework

Abstract

The Good was the 0/0 of self and other (Ch.60); the Beautiful of order and surprise (Ch.62). The True completes the triad as the 0/0 of statement and world. Measured: (1) Twenty Questions - each refutation halves the hypothesis space, from 2^{20} worlds to 1, entropy 20 bits to 0 (Popper 1934); (2) Bayesian concentration (Bayes 1763; Laplace 1774) - a wrong prior rises to the truth $0.35 \rightarrow 0.656$ as variance collapses $1.9e-3 \rightarrow 5.6e-5$; certainty is the zero of doubt; (3) shortest description (Solomonoff 1964) - the lawful data $7n+3$ compress to ratio 0.0002 with a 4-byte law vs 0.355 zlib and 0.659 shuffled: know the law and the description vanishes. Beauty is truth, truth beauty (Keats 1819): compression is the scent of the world-compression we call natural law.

1. Falsification and Twenty Questions (Popper 1934)

$$\log_2(2^{20}) = 20 \text{ bits} \rightarrow 0 \text{ bits}$$

Truth is found by elimination: each query halves the hypothesis set. Measured: 1,048,576 hypotheses left at query 0; 32,768 at 5; 1,024 at 10; 32 at 15; exactly 1 at 20. A single refutation removes half the space - falsification, not verification.

2. Bayesian Concentration (Bayes 1763; Laplace 1774)

$$\text{variance: } 1.90e-3 \rightarrow 5.61e-5 \quad (\text{certainty} = 0 \text{ of doubt})$$

Data Bernoulli(0.65). Prior A starts at the truth (mean 0.65); prior B the student (0.35). Measured posterior means of B: 0.642, 0.639, 0.657, 0.656 - rising to the truth as the variance collapses. Truth emerges as variance goes to 0 - the 0/0 of belief.

3. Truth as Shortest Description (Solomonoff 1964)

$$\text{law-coder ratio } 0.0002 \text{ vs } \text{zlib } 0.355 \text{ vs } \text{shuffled } 0.659$$

The law is the shortest program describing the data. The sequence $7n+3$ (20,000 bytes packed) compresses to a 4-byte rule (first, delta): ratio 0.0002. General-purpose zlib reaches 0.355; the same data shuffled (no law) resists at 0.659. Truth is the code that makes the data compress - the laws of physics are the shortest description of all data, and beauty's compression (Ch.62) is truth's scent.

4. The 0/0 Proof

A statement is nothing by itself - a string. Truth is its correlation with the world - the relation. Meaning was the 0/0 of role (Ch.61); truth is the 0/0 of correspondence. At the boundary description and world coincide: the shortest description is the pattern bared.

5. The Triad (Plato; Kant; Keats 1819)

The Good = 0/0 of self and other (Ch.60)

The True = 0/0 of statement and world (Ch.63)

The Beautiful = 0/0 of order and surprise (Ch.62)

Beauty is truth, truth beauty - that is all ye know on earth, and all ye need to know. (Keats)

6. Connections

Concept	Prior Chapter	Connection
Beauty	Ch.62 Beauty	Compression is truth's scent
Meaning	Ch.61 Meaning	Use becomes correspondence
Suffering	Ch.59 Problem of Evil	Wrong models pay surprise
The cave	Plato (Republic VII)	Shadows are data; sun is law
First cause	Ch.58 First Cause	Axioms: unprovable truth

7. Conclusion

Truth is the 0/0 of the true: the statement is nothing in itself, its truth is the world, and the shortest description is the whole. Popper splits hypothesis space in half per test; Bayes concentrates belief to certainty (variance to 0); Solomonoff finds the 4-byte law that hallucinates 20,000 bytes. Good, True, Beautiful - one removable singularity with three faces: self and other, statement and world, order and surprise.

References

- [1] Popper (1934/1959), The Logic of Scientific Discovery
- [2] Bayes (1763), TRIAL; Laplace (1774), Memoire sur la probabilite
- [3] Solomonoff (1964), A formal theory of inductive inference
- [4] Kolmogorov (1963), Combinatorial foundations of information theory
- [5] Tarski (1933), The concept of truth in formalized languages
- [6] Plato (c. 375 BC), Republic VII (the allegory of the cave)
- [7] Puno, "The Removable Singularity" (2026)